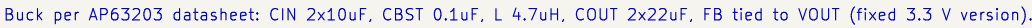
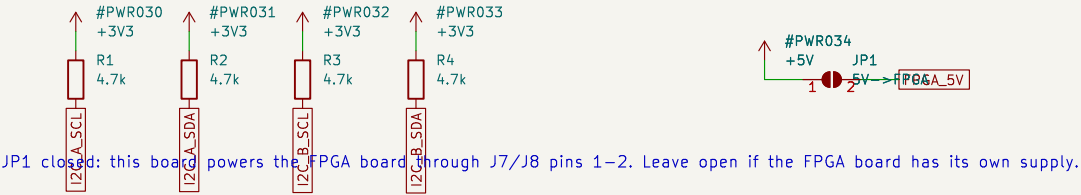
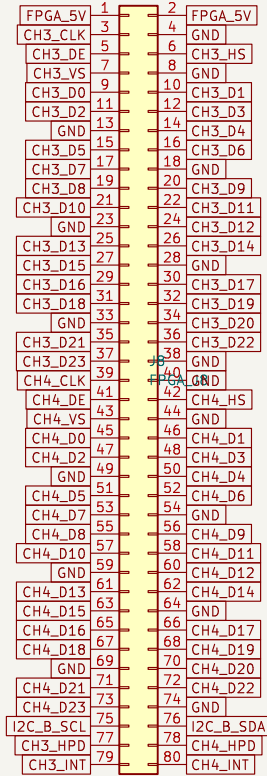
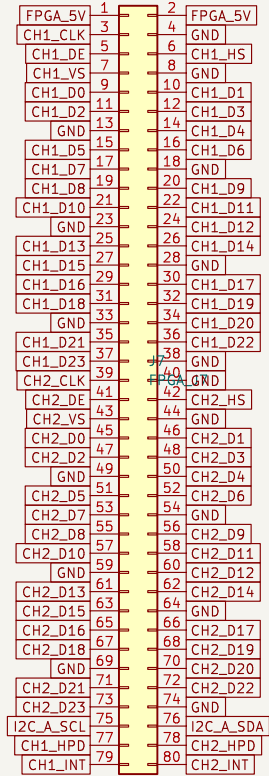


Alternative 5 V input: J6 (2.54 mm header).



Id: 2/

FPGA interface: 2x 80-pin 2.54 mm headers (J7 = OUT1/OUT2, J8 = OUT3/OUT4). 3.3 V LVCMOS. Keep stack height short (< 30 mm) for the 148.5 MHz buses.



JP1 closed: this board powers the FPGA board through J7/J8 pins 1–2. Leave open if the FPGA board has its own supply.

Quad HDMI test pattern generator – ADV7513 x4 carrier

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Sheet: /FPGA interface headers/

File: fpga_conn.kicad_sch

Title: FPGA interface headers

Size: A3

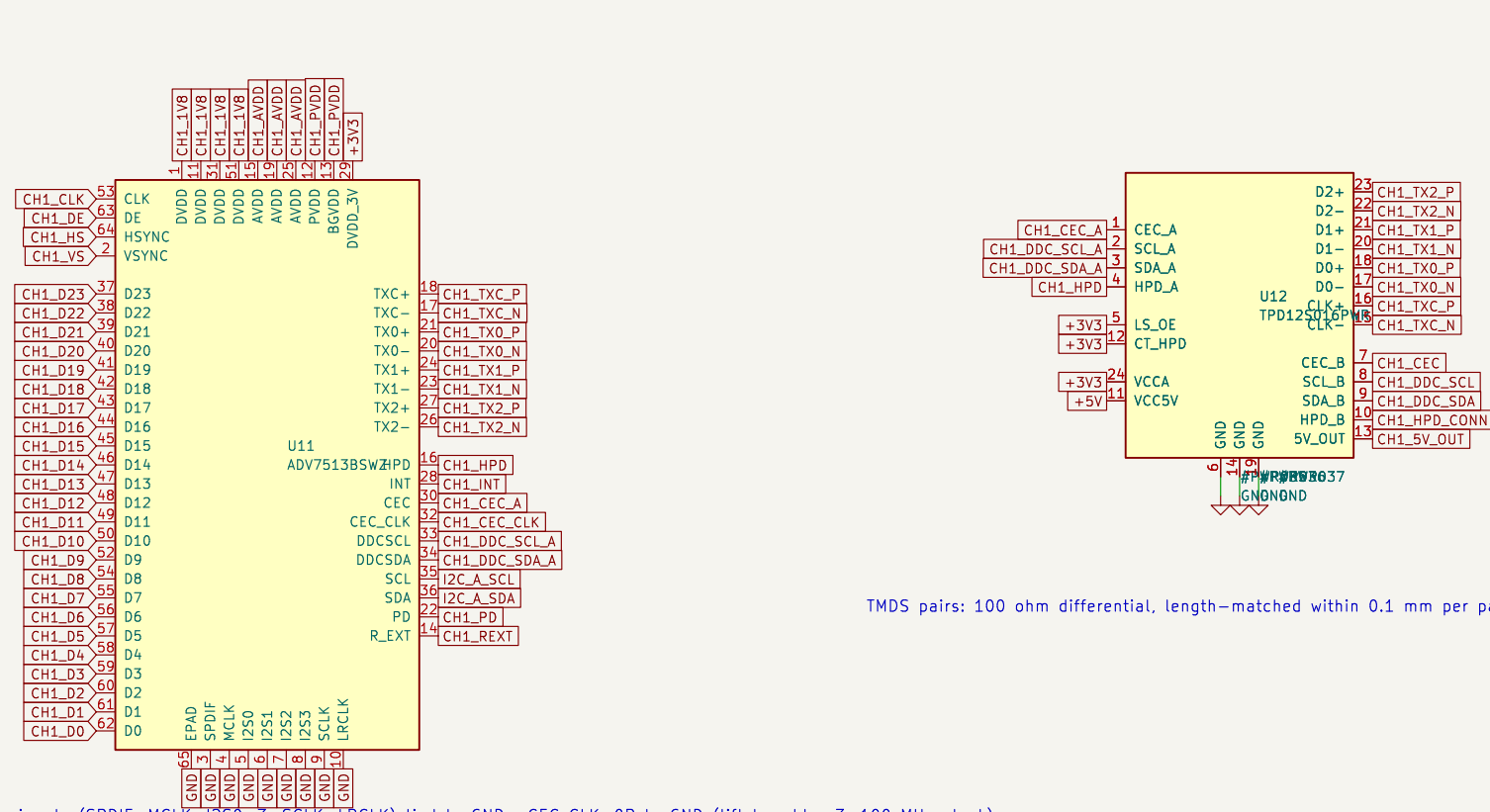
Date: 2026-09-14

Rev: A

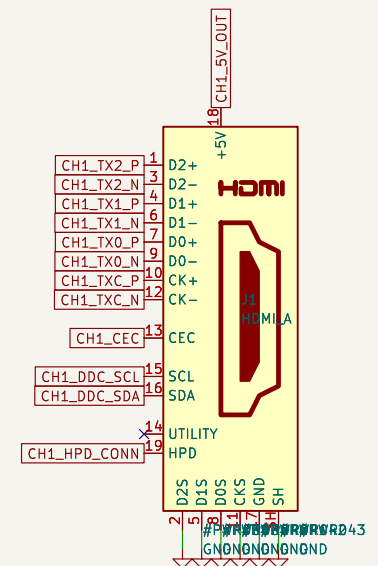
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 3/7

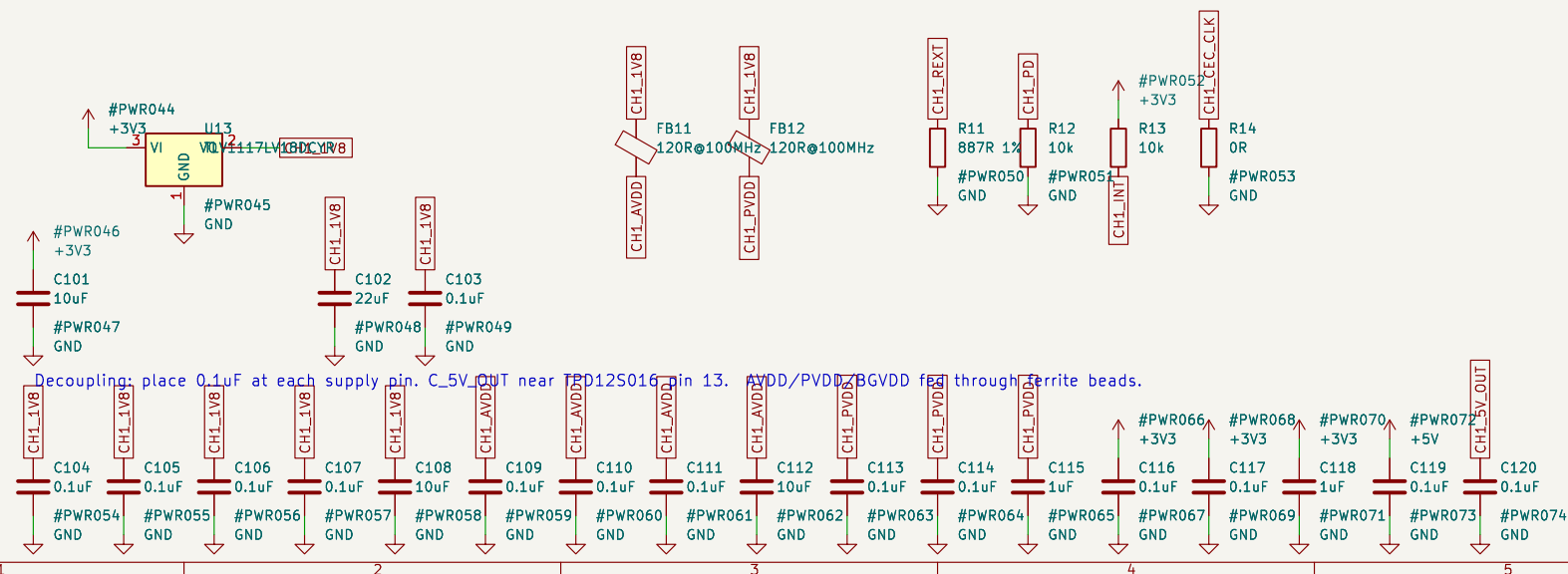
HDMI OUT1 I2C bus A, ADV7513 address 0x72 (PD/AD=L)



TMD5 pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.



Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).



Quad HDMI test pattern generator – ADV7513 x4 carrier

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Sheet: /HDMI OUT1: ADV7513 + TPD12S016/

File: hdmi_ch1.kicad_sch

Title: HDMI OUT1: ADV7513 + TPD12S016

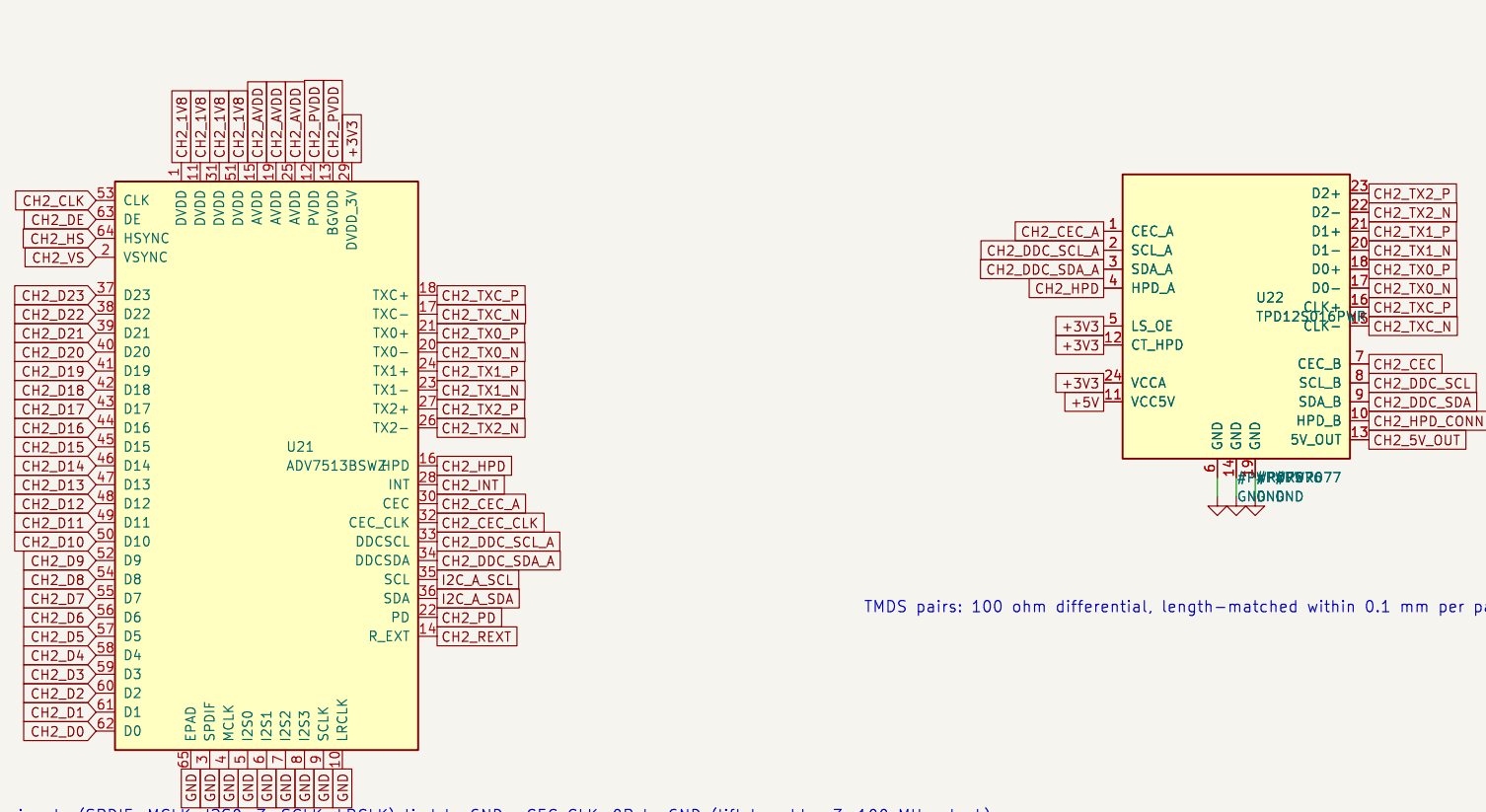
Size: A3

Date: 2026-09-14

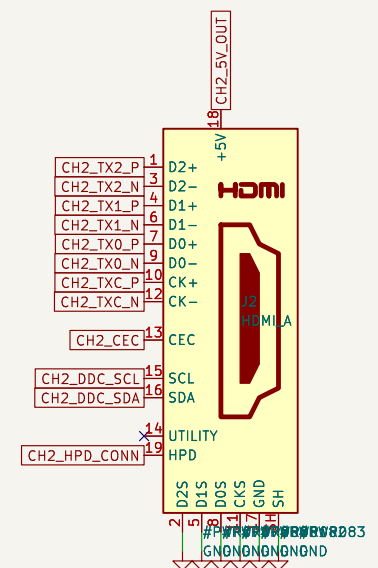
Rev: A

Size: A5	Date: 2020-09-14
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	

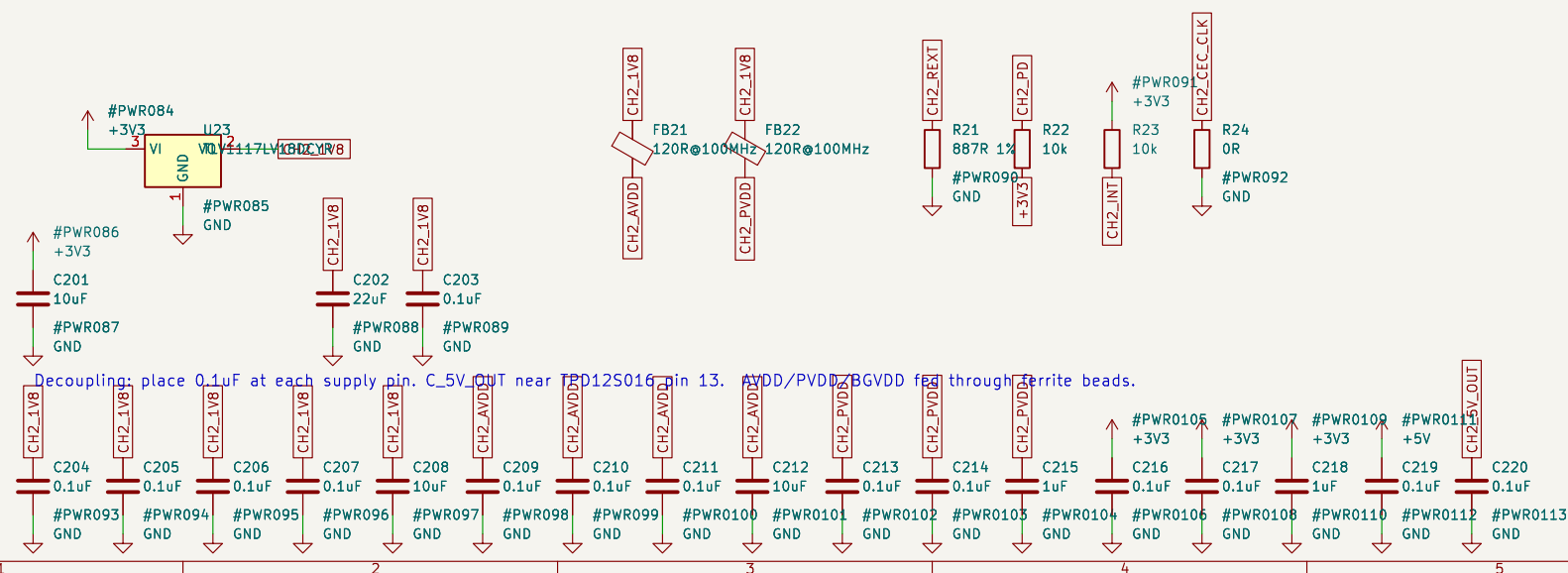
HDMI OUT2 I2C bus A, ADV7513 address 0x7A (PD/AD=H)



TMD5 pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.



Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: 0R to GND (lift to add a 3-100 MHz clock).



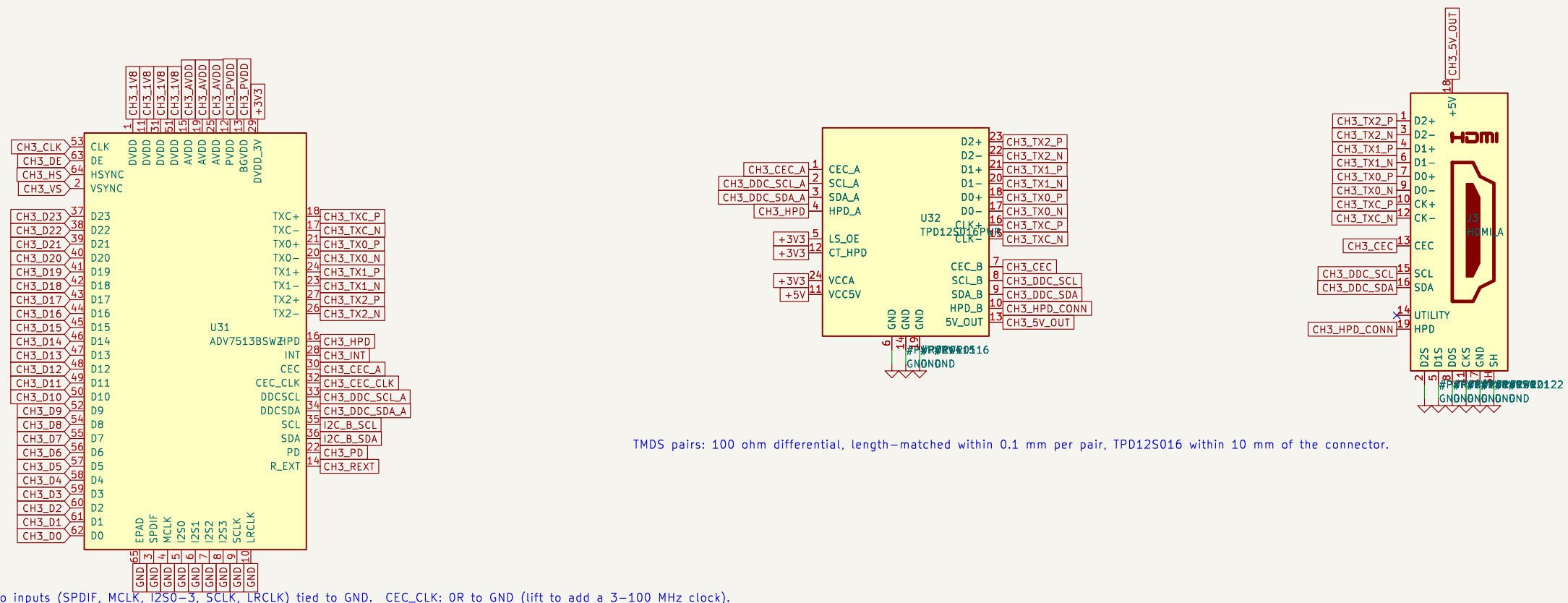
Quad HDMI test pattern generator – ADV7513 x4 carrier
gyuhooo

Sheet: /HDMI OUT2: ADV7513 + TPD12S016/
File: hdmi_ch2.kicad_sch

Title: HDMI OUT2: ADV7513 + TPD12S016

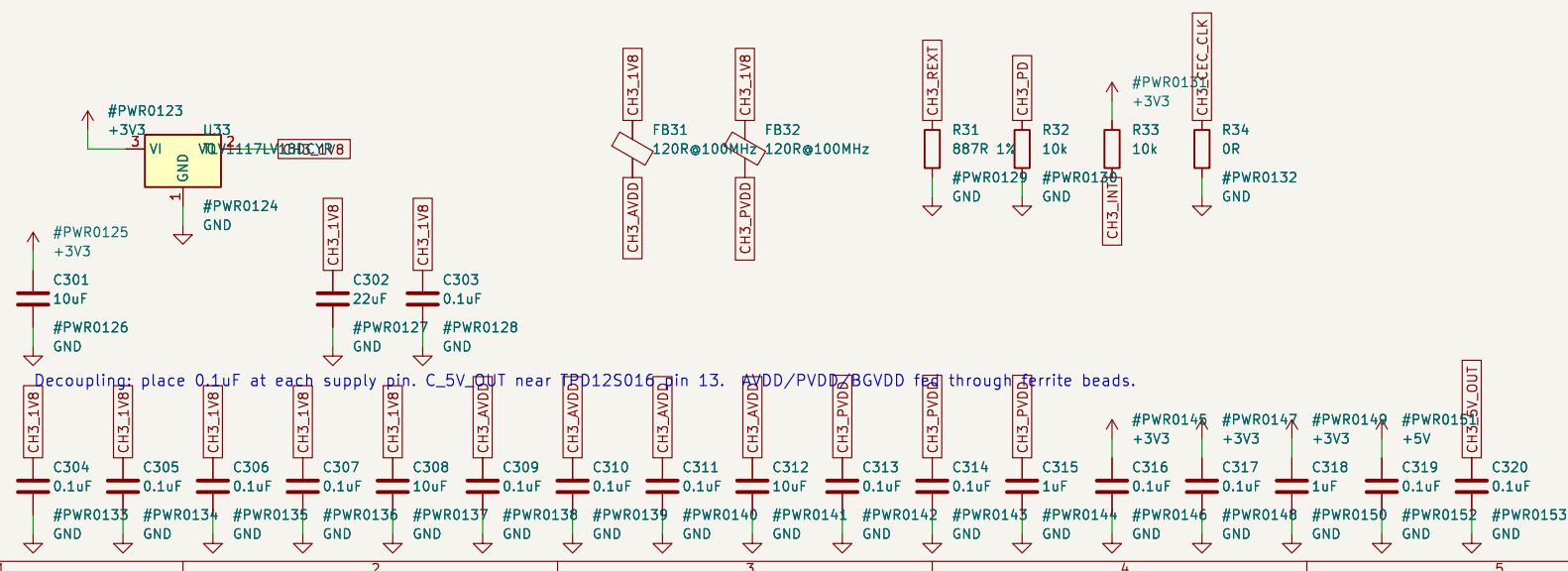
Size: A3	Date: 2026-09-14	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/7

HDMI OUT3 I2C bus B, ADV7513 address 0x72 (PD/AD=L)



Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: 0R to GND (lift to add a 3-100 MHz clock).

TMD5 pairs: 100 ohm differential, length-matched within 0.1 mm per pair, TPD12S016 within 10 mm of the connector.



Quad HDMI test pattern generator – ADV7513 x4 carrier

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Sheet: /HDMI OUT3: ADV7513 + TPD12S016/

File: hdmi_ch3.kicad_sch

Title: HDMI OUT3: ADV7513 + TPD12S016

Size: A3

Date: 2026-09-14

Rev: A

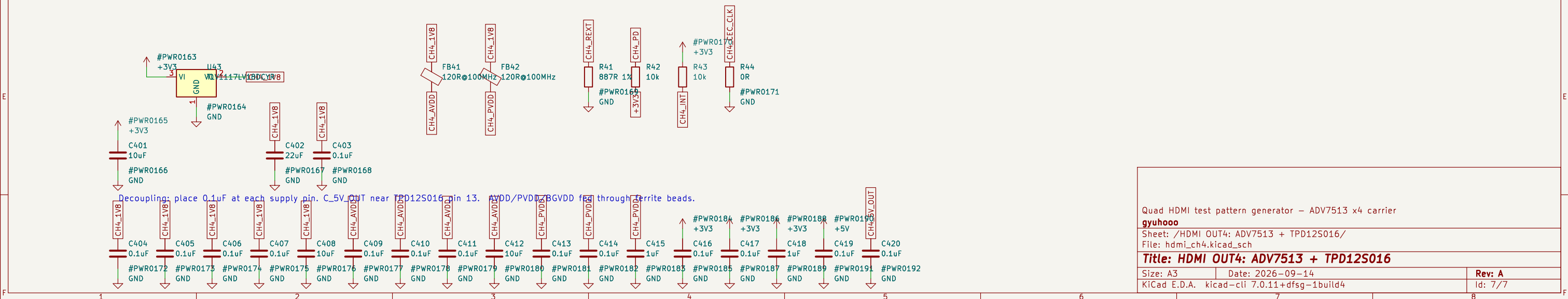
Size: A3	Date: 2020-09-14
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	

REV: A

Ed: 6/7

CH4_D7 59 D7 SDA 58 12C_B_SDA
CH4_D6 56 D6 SCL 57 12C_B_SCL

Unused audio inputs (SPDIF, MCLK, I2S0-3, SCLK, LRCLK) tied to GND. CEC_CLK: OR to GND (lift to add a 3-100 MHz clock).



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File: hdmi_ch4.kicad_sch

Size: A3	Date: 2026-09-14	Rev: A
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KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 7/7

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